

RESEARCH ARTICLE

An In-Depth Investigation of Genetic Programming Under Physical Time and Directional Change Frameworks for Algorithmic Trading

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ABSTRACT The majority of algorithmic trading studies rely on datasets with fixed physical time intervals, such as hourly or daily data, resulting in a discontinuous representation of time. Directional Changes (DC) is an alternative method, which transforms these datasets from physical time (PT) intervals into event-driven sequences, allowing for unique price analysis. We introduce two novel algorithms based on Genetic Programming (GP): GP-DC, which only uses DC-based indicators as features, and GP-DC-PT, which combines DC-based indicators with physical-time indicators from technical analysis to create trading strategies. We extensively evaluate the return and risk of DC-based trading strategies in 220 datasets from ten international financial markets for periods up to ten years. We compare the performance of these trading algorithms against a non-DC-based GP, technical analysis derived strategies, as well as the well-known benchmark of buy-and-hold. Our results demonstrate that the proposed DC-based algorithm GP-DC creates profitable trading strategies with low risk, and statistically and significantly outperforms the other benchmarks. We also find that the feature set that includes both DC and physical time indicators (GP-DC-PT) leads to further improvements in algorithmic trading performance and is very competitive, as it achieves an average total return of over 18%. These findings suggest that including directional changes in trading strategies significantly enhances their return and risk performance.

INDEX TERMS Algorithmic trading, directional changes, genetic programming, stock market.

I. INTRODUCTION

Algorithmic trading has become a focal point of investigation within the financial domain [1]. Most research in algorithmic trading traditionally relies on fixed physical time scales, such as hourly, daily, or weekly intervals. However, this conventional approach often leads to discontinuous data, resulting in the loss of significant information between data points. For example, if one uses daily closing prices, they would obtain only a single price point per day, ignoring all price changes that occur during the day. All these ‘ignored’ data points are often missed opportunities, both

from a financial perspective (e.g., a trading algorithm could take advantage of these intraday price changes and make significant profit) and from a machine learning perspective, due to the fact that these additional data could have been used for training an algorithm. To avoid this issue, one could start using high-frequency data, such as hourly, or even at a higher minute-by-minute. However, the same problem occurs (albeit on a smaller scale): summarising data under fixed time intervals ignores what happens in between them. Alternatively, one could use tick-by-tick data, i.e., record every individual price change. While this offers the most accurate and detailed view of market activity, it introduces its own challenges, such as large amount of data and cost, as acquiring and maintaining tick-by-tick

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data can be expensive due to its high frequency and volume.

To avoid these challenges that derive from the use of physical time, one could represent price movements as discrete events, effectively capturing noteworthy price fluctuations, typically defined by thresholds, e.g., a 5% change. As a result, data is summarised by capturing significant market activities. The framework of Directional Changes (DC), introduced in [2] and formalised in [3], leverages a user-defined threshold value, denoted as θ , to detect substantial price movements. In the DC framework, a price series is partitioned into distinct upward and downward trends, marked by DC events when prices deviate beyond θ . These DC events are then followed by overshoot events, signifying price movements beyond the DC event. Therefore, directional changes emphasize the magnitude of price changes over varying time intervals, unlike the traditional approach under physical time which uses fixed intervals (e.g., daily closing prices). This concept offers traders a fresh perspective for analyzing price movements, enabling them to concentrate on significant price changes while filtering out less relevant price details.

In this article, we propose combining DC-based indicators (features) with more traditional indicators derived from physical time, namely technical analysis. We aim to show that these two families of indicators have complementary characteristics, and are thus able to attain significantly improved performance when considered together. To achieve this, we use a genetic programming (GP) algorithm, which is a well-known and well-documented algorithm in the field of algorithmic trading [1]. The GP will allow both families of indicators to be considered together when creating trading strategies. We present an overview of genetic programming in Section II.

To assess the performance of the GP trading algorithm, we conduct a comprehensive analysis across 110 datasets from ten international markets spanning six different countries. The investigation covers two distinct time periods, spanning 5 and 10 years, totaling over 220 datasets. The primary aim is to explore the potential of DC while demonstrating its competitiveness and capacity to outperform the traditional physical time paradigm. To benchmark the DC-based algorithms, we compare them against a GP-based strategy, namely GP-PT, which uses traditional physical time technical analysis indicators. Performance evaluation is based on a range of financial metrics, including total return, rate of return per trade, risk, and the Sharpe ratio. Additionally, the GP-based algorithms are compared to three popular technical analysis indicators and the buy-and-hold strategy.

The rest of this article is organised as follows. Section II provides essential background information on directional changes and genetic programming, and also presents the relevant literature. Section III introduces the methodology and presents our novel GP that incorporates DC-based indicators. Section IV details the experimental setup, introducing the datasets, benchmark strategies, and parameters' tuning.

Then, we discuss our experimental findings in Section V, and, finally, Section VI concludes this work and discusses potential future directions.

II. BACKGROUND AND LITERATURE REVIEW

This section provides a brief background of DC and GP together with related work on these topics.

A. DIRECTIONAL CHANGES

Directional Changes (DC) offers a different way to understand market price movements compared to the usual fixed-interval method. With DC, an event occurs when a financial asset's price movement goes beyond a certain threshold set by traders. These events are categorized as either upturn or downturn events depending on the price direction. Recent studies, however, have questioned whether every DC event must lead to an overshoot (OS) event [4].

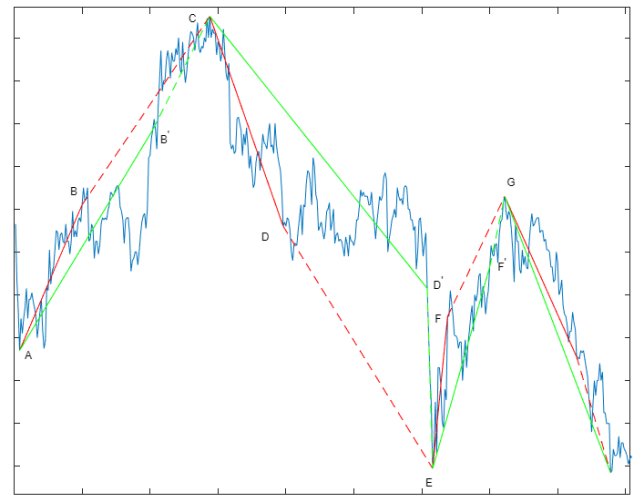


FIGURE 1. An example of the DC framework, presenting the physical time series (blue line), two series of DC and OS events as defined by a threshold of 1% (red line) and 1.5% (green line). We use solid lines for DC and dotted lines for OS events.

Figure 1 shows how we convert a continuous price series into DC and OS events under two different thresholds (see red and green lines). It is crucial to know that traders can choose these thresholds, and this choice affects the events identified. Lower thresholds result in more events, creating more trading opportunities, while higher thresholds give fewer events with more significant price changes. So, selecting the right threshold sets an interesting challenge.

When examining the historical daily price data, represented by the blue line, and events detected using a 1% threshold, indicated by the red lines, we observe that many price fluctuations do not meet the criteria for being considered events. An event is only triggered when the price movement crosses this threshold, as indicated by the dashed lines.

In essence, connecting point A to point B signifies an upward trend, followed by a downturn from point B to point C, represented by solid red lines. It's crucial to emphasize that

we can only confirm a change in the trend when the price surpasses the specified threshold. In other words, we must wait to confirm the next event in the opposite direction before recognizing the end of an ongoing event.

For instance, in Figure 1, point D acts as a confirmation point for a DC event. Before reaching point D, the previous OS event is still active, and traders interpret it as an upward trend. This presents a challenge: for optimal returns, it is ideal for trades to be closed as near as possible to the conclusion of the OS event. However, by the time this becomes apparent, it's often too late. Therefore, determining turning points, such as point C in Figure 1, remains a continuous area of research in the field of DC, with various methods proposed to estimate the duration of OS events [5].

The advantage of DC is that it provides traders with a fresh perspective on price movements, helping them concentrate on significant events while filtering out other price fluctuations that could be just noise. As a result, DC presents new research opportunities and challenges not found in traditional time intervals, as discussed in the existing literature in Section II-C.

B. GENETIC PROGRAMMING

Genetic Programming (GP) was originally introduced as a distinct research technique by Koza [6] and has found applications in financial forecasting for more than two decades [1]. For instance, Manahov et al. [7] developed a GP-based 1-day-ahead trading system that allowed for the exploration of stocks through artificial traders' groups, demonstrating the system's superiority over benchmarks for such short-term prediction.

In addition, Becker and Seshadri [8] introduced GP-based technical trading rules whose performance surpassed that of the buy-and-hold strategy for the S&P500, even when accounting for transaction costs. GP's effectiveness was also compared with neural networks, another popular machine learning model, as reported by Iba et al. [9]. Pimenta et al. [10] developed an automated system integrating multi-objective optimization, genetic programming, and technical analysis. The system's performance was evaluated with six BOVESPA shares during two periods (2/2013 - 2/2015 and 7/2015 - 7/2016).

Graff et al. [11] provided evidence that GP-based systems compete with traditional algorithms, such as ARIMA and exponential smoothing, and in some cases, GP exhibited statistically superior performance. Besides, Long et al. [12] conducted a comprehensive investigation into 10 machine learning algorithms, including GP, across the performance of different stocks in 10 markets. The results demonstrated GP's outperformance relative to other algorithms. Ari and Alagoz [13] applied a hyperparameter optimal genetic programming-based forecast model evaluated for the one-day-ahead prediction of the stock market. Their experiments showed 4.8% more average income than a long-term investment strategy.

C. RELATED WORK

Directional Changes (DC), originally introduced by Guillaume et al. [2] and defined in a formal manner by Tsang [3], offer an event-driven alternative to the conventional fixed-time model. DC has gained traction in handling non-constant time intervals and high-frequency data, notably in tick data from the Forex market, as evidenced in works such as [14], [15], [16].

Within the DC framework, two critical concerns emerge. Firstly, the determination of overshoot (OS) event end-points holds significant importance for maximizing profits. Glattfelder et al. [5] defined empirical scaling laws to quantitatively connect price movements and transactions in the foreign exchange market. Building on this work, Aloud and Fasli [17] extended it with additional scaling laws, yielding positive outcomes in the foreign exchange market. One notable scaling law indicates that, on average, an OS event takes twice as long as the DC event to achieve the same price change. Adegboye and Kampouridis [4] put forward a novel DC trading strategy that challenges the assumption that DC events are always succeeded by OS events, demonstrating superior performance compared to other DC-based strategies and the buy-and-hold approach across 20 Forex currency pairs. Rostamian and O'Hara [18] used the Average True Range to determine the threshold of the DC and suggested that CNN-LSTM network architecture achieves improved performance. In a recent study, Adegboye et al. [19] utilized genetic algorithms to optimize multiple DC-based trading strategies using data from 20 foreign exchange markets, outperforming single DC-based strategies, traditional technical indicators, and the buy-and-hold approach.

The second significant issue revolves around the integration of technical analysis within the DC framework. Traditionally, technical analysis has been applied in physical time, primarily employing technical indicators. Aloud [20] introduced the first set of DC-specific indicators, with additional DC indicators proposed in [21] and [22]. Ao and Tsang [23] introduced DC-based trading strategies resulting in positive returns across various stock market indices. More recently, Aloud and Alkhamees [24] used reinforcement learning as part of a DC framework for trading in S&P500, NASDAQ, and Dow Jones stock markets, achieving substantial returns and an improved Sharpe ratio.

From the above it is evident that there is relatively limited research on the application of DC in the stock market. Furthermore, exploiting indicators from the DC paradigm for constructing trading strategies is still largely unexplored compared to the technical analysis in the physical time domain. One notable exception to the above is our previous work [25], which was the first to introduce DC indicators within a GP framework. However, while it yielded promising results, it was tested on a limited dataset of 33 stocks from 3 international markets. In our current research, we will conduct a more extensive experimental study on 220 stocks from 10 international markets. Additionally, we will evaluate

the algorithmic performance under different circumstances, such as different markets and time periods, and consider additional evaluation metrics, including total return, beyond those used in [25]. More importantly, this paper will introduce and evaluate a more sophisticated GP model that combines DC indicators with physical time indicators (GP-DC-PT), expanding on the GP-DC model introduced previously.

Given the demonstrated effectiveness of GP in combining diverse indicators for profitable trading strategies, as observed in [26], [27], [28], and [29] and the promising results of transformer-based models using technical indicators [30], [31], it is natural to want to further explore the potential of using DC-based indicators as part of a GP trading algorithm. In addition, we are interested in comparing the aforementioned GP-DC-PT with algorithms that use technical analysis indicators based on the physical time paradigm. In the subsequent sections, we introduce the GP methodology and present the trading strategy we have employed.

III. METHODOLOGY

We propose two new GP algorithms that take advantage of the DC framework. The first algorithm, which we call GP-DC, is a GP algorithm that solely employs trading strategies using Directional Changes (DC)-based indicators. The second algorithm, termed GP-DC-PT, is a GP algorithm that uses both DC-based and physical time indicators. For completeness, we also run experiments with a third GP algorithm, namely GP-PT, that only uses physical time indicators and acts as a benchmark to the two proposed algorithms mentioned above. We call this third algorithm GP-PT.

For the rest of this section, we first detail the terminal set indicators used on each GP algorithm. Then, we continue presenting other relevant GP components, such as the function set, model representation, fitness function, as well as selection method and operators, which are the same for all GP algorithms. Lastly, we introduce the trading strategy, which is again used by all three GPs, as well as the fitness function.

A. GENETIC PROGRAMMING SETUP

1) TERMINAL SET

We start by collecting daily closing prices from historical datasets. These prices are then processed using the Directional Changes (DC) framework to distil them into significant events. We subsequently calculate 28 specific DC indicators, which have been previously introduced and discussed in [20]. These DC indicators are conceptually similar to the technical indicators used in traditional time-based technical analysis, as detailed in [32]. In conjunction with these indicators, we employ an Ephemeral Random Constant (ERC), which, upon invocation, generates a random number distributed uniformly between -1 and 1 . To ensure consistency with ERC's range, the DC indicators have been appropriately normalized.

Table 1 provides a comprehensive listing of these 28 DC indicators. Notably, among these indicators, 10 of them involve calculations over specific time intervals, such as the total count of DC events denoted as N_{DC} , which can be computed over varying time windows, including 10, 20, 30, 40, or 50 days. This brings the total count of indicators to 28. For indicators that do not depend on a specific time window, such as OSV, TMV, T_{DC} , and R_{DC} , Table 1 designates them as "N/A."

TABLE 1. DC indicators; see also [20].

Indicator	Description	Periods (days)
TMV	Price change between the extreme points defining the beginning and end of a trend, normalised by the threshold θ .	N/A
OSV	% difference of current price and the last DC confirmation point, divided by the threshold θ .	N/A
Average OSV	Average value of the OSV indicator over a given period.	3, 5, 10
R_{DC}	Time-adjusted return of DC. It is equal to the product of TMV and the threshold θ , divided by the time intervals between each extreme point.	N/A
Average R_{DC}	Average value of the R_{DC} indicator over the given period.	3, 5, 10
T_{DC}	Duration of a DC trend.	N/A
Average T_{DC}	Average value of the T_{DC} indicator over the given period.	3, 5, 10
N_{DC}	Total number of DC events over the given period.	10, 20, 30, 40, 50
C_{DC}	Sum of the absolute value of the TMV indicator over the given period.	10, 20, 30, 40, 50
A_T	The disparity between the time DC classifies as up trends and down trends over the given period.	10, 20, 30, 40, 50

In addition to the above 28 DC-based indicators, we also use 28 physical time indicators based on technical analysis. Some of these indicators (MA, CCI, RSI, and William's %R) are considered for periods of 10, 20, 30, 40, and 50 days, a couple of them (ATR and EMA) are considered for periods of 3, 5, and 10 days, while finally, OBV and PSAR do not involve periods [33]. Table 2 summarises these indicators.

We employed Python's TA-Lib to compute the 28 physical-time indicators, whereas the 28 DC-based indicators were calculated following the formulas proposed by [20].

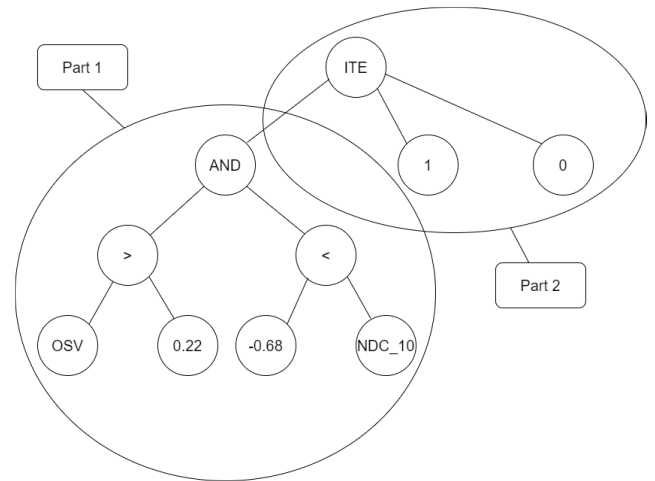
The proposed GP-DC algorithm uses only DC indicators in its terminal set, thus only the indicators presented in Table 1. In addition, the proposed GP-DC-PT algorithm uses a combination of DC and physical time indicators, as presented in Tables 1 and 2. Lastly, the benchmark GP-PT uses only physical time technical analysis indicators, thus the indicators presented in Table 2.

2) FUNCTION SET

We consider two logical operators, AND and OR, as well as two logical expressions, i.e., less than ($<$) and greater than ($>$). This allows us to simulate actions of actual traders by comparing technical indicators to appropriate thresholds.

TABLE 2. Physical time (technical analysis) indicators; see also [33].

Indicator	Description	Periods (days)
MA	Moving average for a given period.	10, 20, 30, 40, 50
CCI	Commodity Channel Index, which measures the deviation of an asset's price from its statistical average.	10, 20, 30, 40, 50
RSI	Relative Strength Index, which is a momentum oscillator to measure the magnitude of recent price changes and determine overbought or oversold conditions of an asset.	10, 20, 30, 40, 50
William's %R	Measures oversold or overbought conditions of an asset by comparing the closing price of an asset to its price range over a set period of time.	10, 20, 30, 40, 50
ATR	Average True Range, which measures the volatility of an asset by calculating the average of the true range over a set period of time.	3, 5, 10
EMA	Exponential Moving Average, which calculates a weighted average of a series of prices over a set period of time, where more recent prices are given greater weight in the calculation.	3, 5, 10
OBV	On Balance Volume, which measures buying and selling pressure, by calculating the cumulative total of an asset's volume, where positive volume is added to the total of an up day and negative volume is subtracted on a down day.	N/A
PSAR	Parabolic Stop and Reverse, which identifies potential reversals in the direction of an asset's price movement by placing dots on a chart that indicate potential stop and reverse points for a long or short position.	N/A


FIGURE 2. A GP tree: if OSV is greater than 0.22 and N_{DC} for 10 days is greater than -0.68 , the 'Then' branch suggests a buy action, otherwise the 'Else' branch suggests we hold. The thresholds 0.22 and -0.68 are two random values generated by ERC.
TABLE 3. Configuration of the GP algorithm.

Configuration	Value
Function set	AND, OR, $>$, $<$
Terminal set	28 DC indicators, ERC, and 28 technical indicators
Genetic operators	Elitism, subtree crossover and point mutation
Selection	Tournament

3) MODEL REPRESENTATION

The GP evolves logical expressions using operators like AND, OR, $<$, or $>$. These expressions become the initial branch of an If-Then-Else (ITE) statement, as shown in Part 1 of Figure 2. The ITE tree includes also a 'Then' branch, which stands for a buy action and always returns a value of 1, and an 'Else' branch, which represents a hold action and consistently returns a value of 0. Note that this structure does not cover a sell action, and we will provide more details on that in Section III-B. Part 2 of the GP is not involved in evolution because it deals with constant values (0 or 1). The choice between the 'Then' branch (indicating a buy signal) and the 'Else' branch (indicating a hold signal) depends on whether the GP individual's expression evaluates to True or False.

4) SELECTION METHOD AND OPERATORS

In our computational framework, we employ elitism, subtree crossover, and point mutation as genetic operators. Furthermore, we utilize tournament selection to determine the parental candidates for the aforementioned genetic operators [34]. An overview of the GP configuration is provided in Table 3.

B. TRADING STRATEGY

The objective of a GP individual, representing a trading strategy, is to predict if the price will increase by a given percentage, denoted as $r\%$, within a predefined time frame of n days. If the GP individual generates a "True" prediction,

we initiate a purchase of a single unit of the stock unless we currently hold it. Conversely, if the prediction is "False", we maintain our current position (hold). In instances where we already possess the stock and its price rises by at least $r\%$ within the subsequent n days, we execute a sale on the day of the increase. If the price fails to increase by the defined threshold within the specified time frame, we sell the stock on the n -th day. We remark that our trading strategy does not involve short-selling. After each selling action, we compute the resulting profit, incorporating transaction costs set at 0.025% per trade. By incorporating transaction costs into the evaluation process, the resulting profit may be reduced. However, this adjustment is crucial as it better reflects the conditions of real-world trading. In addition, including transaction costs discourages the algorithm from exploiting unrealistic high-frequency behaviour that is unlikely to be feasible in the real market. Instead, it encourages the discovery of more robust trading strategies. The summary of our trading strategy is outlined in Algorithm 1.

To assess the trading strategy's performance for a specific stock, we rely on four key metrics: Rate of Return (RoR), Total Return (TR), risk, and the Sharpe ratio. RoR, calculated for each trade (i.e., i), hinges on the purchase price $P_b(i)$ and selling price $P(i)$, while accounting for transaction costs. The returns are preserved in a list, labeled RoR. For an overall evaluation, we aggregate the returns from this list, dividing the sum by the initial cost incurred in the first transaction to derive the Total Return, as defined in Equation 2. Both RoR

Algorithm 1 Our Trading Strategy With Input the Threshold $r\%$ and the Number of Days n

Require: O is the outcome of the GP tree and $flag \in \{0, 1\}$ indicates whether we already hold the stock (i.e., 1) or not (i.e., 0)

```

1: if  $O = 1$  and  $flag = 0$  then
2:   Buy one share of stock and set  $flag \leftarrow 1$ 
3:    $t_0 \leftarrow t$  //Mark the time of trade:  $t$  is used for the
   current time
4:    $p_0 \leftarrow p$  //Mart the price when buying:  $p$  is used
   for the current price
5: else
6:   if ( $flag = 1$  and  $p > (1+r/100) \times p_0$ ) OR  $(t-t_0) > n$ 
   then
7:     Close the trade and set  $flag \leftarrow 0$ 
8:     Calculate the resulting profit
9:   end if
10: end if

```

and TR provide insights into trade-level profitability and the total returns over the entire trading period.

The third metric, risk, is quantified as the standard deviation of the RoR list, as expressed in Equation 3. Finally, our evaluation extends to the Sharpe ratio, which unifies returns and risk considerations, detailed in Section III-C.

$$RoR(i) = \frac{(1-c) \cdot P(i) - (1+c) \cdot P_b(i)}{(1+c) \cdot P_b(i)} \cdot 100\% \quad (1)$$

$$TR = \frac{\sum_{i \in RoR} (1-c) \cdot P(i) - (1+c) \cdot P_b(i)}{(1+c) \cdot P_b(1)} \cdot 100\% \quad (2)$$

$$Risk = \sqrt{Var(RoR)} \quad (3)$$

C. FITNESS FUNCTION

We use the Sharpe ratio, defined in Equation 4, as the fitness function for all three GP algorithms. That is, each algorithm's objective is to maximize the Sharpe ratio, which takes into account both returns and risk.

$$Sharpe\ ratio = \frac{E(RoR) - R_f}{Risk}, \quad (4)$$

where $E(RoR)$ stands for the sample mean of the list of the rate of returns for a given stock and R_f is the risk-free rate.

In this way, we optimize trading strategies based on a comprehensive metric that considers both returns and risk, rather than focusing solely on one aspect.

IV. EXPERIMENTAL SETUP

In this section, we introduce the financial data we used and we present the benchmarks for the experimental comparison. Furthermore, we discuss the parameter-tuning process for the GP and the trading strategy. Finally, we present a detailed description of the research procedure.

A. DATA

To assess the performance of Genetic Programming (GP) algorithms utilizing technical indicators derived from the DC framework in the context of the stock market, we gathered daily price data from 10 randomly selected stocks from each of 10 international markets spanning 6 countries in Europe (Financial Times Stock Exchange 100 Index, the DAX performance index), USA (Dow Jones Industrial Average, Nasdaq Stock Market, New York Stock Exchange, Russell 2000 Index, Standard and Poor's 500), and Asia (Nifty Fifty, Taiwan Stock Exchange Corporation, Nikkei 225).

From each market, we obtained two datasets covering a 5-year period (from 24/11/2015 to 24/11/2020) containing around 1250 data points and a 10-year period (from 25/11/2010 to 24/11/2020) containing around 2500 data points for 100 stocks, sourced from Yahoo! Finance. Additionally, we collected data for the market indices themselves. In total, this amounted to 220 datasets, considering both the 10 markets, 10 stocks per market, and two time periods.

Each dataset was subdivided into the corresponding training set (60%), validation set (20%), and test set (20%). We utilized the training and validation sets for optimizing relevant parameters, while the final experiment involved training a combined dataset (comprising the training and validation data) and testing it against the dedicated test set. All the datasets and experimental code are available at "https://github.com/xinpeng255/GP_DC_financial.git".

B. BENCHMARKS

The two novel GP algorithms, namely GP-DC and GP-DC-PT, are benchmarked against GP-PT, which only uses physical time. This allows us to understand the added value of applying GP algorithms that include also DC-based indicators, apart from physical time ones.

Our second benchmark is trading strategies derived from physical time technical analysis indicators. Such indicators are very common in both the financial literature, as well as the trading industry, so we consider it important to compare our proposed algorithms' performance against such indicators. We run trading experiments for the following indicators: Moving Average Convergence/Divergence (MACD), On Balance Volume (OBV), and Momentum (MTM).

Lastly, the final benchmark is the buy-and-hold strategy; a trading strategy where traders buy a stock and hold it for a long time with the hope of an increase in a long period [35]. The net profit in buy-and-hold is the difference between the price at the beginning and the end of the trading period. Buy-and-hold allows us to compare the GP-based algorithms' performance against the market's and, hence, we will thus also report their performance on each dataset.

C. PARAMETER TUNING

We conducted a grid search to determine the optimal GP parameters for three GP-based algorithms, with parameter tuning carried out using the validation set. Table 4 presents

the selected parameters and their corresponding values after tuning.

TABLE 4. Selected parameters of the GP algorithm after parameter tuning.

Parameter	Value
Max depth	6
Population size	500
Crossover probability	0.95
Mutation probability	0.05
Tournament size	2
Numbers of generation	50

For the trading strategy, we need to decide two parameter values from the question, “Will the price increase by at least $r\%$ in the next n days?” to make sense. Additionally, the DC-based GP models (GP-DC and GP-DC-PT) introduce an extra parameter requiring tuning, namely the DC threshold (θ).

Instead of globally tuning these parameters to come up with a best set of values across all datasets (as done for the GP), we opted for customized values for each individual dataset. This approach aimed to maximize the performance of the trading strategy on a per-dataset basis. These three parameters take values from the following space:

- n (prediction horizon): 1, 5, 15
- r (price movement percentage): 1%, 5%, 10%, 20%
- θ (DC threshold): 0.001, 0.002, 0.005, 0.01, 0.02

D. EXPERIMENTAL PROCEDURE

We summarize the step-by-step procedure followed in this study to develop and evaluate the proposed trading strategy.

1) DATA COLLECTION

As introduced in Section IV-A, we select and download the daily price of 10 different stocks from each of 10 international markets on Yahoo! Finance. To ensure a broad and representative analysis, the selected markets span six countries. For each stock, we select historical data covering two timeframes: a 5-year period and a 10-year period, allowing us to examine the impact of the time horizon on financial forecasting performance. Additionally, we collect the corresponding market index data for each market.

2) FEATURE EXTRACTION

To investigate the GP under the physical time and DC framework, we compute 28 physical technical indicators using the Python library `TA-Lib` and 28 DC indicators following the formulas proposed by [20]. These indicators were calculated for each stock across both the 5-year and 10-year timeframes.

3) ALGORITHMIC SETUP

To achieve the aims of this paper, we design and configure three GP-based algorithms (GP-DC, GP-PT, GP-DC-PT) using a shared function set and evolutionary setup,

as described in Section III-A. The key difference among these algorithms is the terminal sets:

- GP-DC: uses only the 28 DC indicators,
- GP-PT: uses only the 28 physical time indicators,
- GP-DC-PT: uses both 28 DC and 28 physical time indicators.

4) TRAINING AND TUNING

To ensure that parameter selection does not bias the experimental results, each dataset is divided into three subsets: a training set (60%), a validation set (20%), and a test set (20%). The entire parameter tuning process is conducted using only the training and validation sets. We perform a grid search for GP parameters and a dataset-specific tuning of the prediction horizon, price movement percentage and DC thresholds, as detailed in IV-C. Following parameter tuning, the algorithms are trained using a combination of the training set and the validation set.

5) EVALUATION

We test the trained models on unseen data (the test set) using total return, rate of return, risk, and Sharpe ratio. The performances of the GP-DC and GP-DC-PT are compared to the GP-PT to investigate the effectiveness of the proposed algorithm under different frameworks. Additionally, the GP-DC-PT is compared with benchmark strategies, including three commonly used technical indicators (MACD, OBV, and MTM) and a buy-and-hold strategy.

6) STATISTICAL ANALYSIS

To support our findings, we use the Kolmogorov-Smirnov test to assess the significance of performance differences. This non-parametric test evaluates whether two distributions differ significantly. We apply the Kolmogorov-Smirnov test to compare the total return, rate of return, and risk across different algorithms and benchmark strategies.

V. RESULTS AND ANALYSIS

The goal of our experiments is to provide evidence that DC has the potential to outperform the physical time paradigm. To achieve this goal, we evaluate GP-DC using GP-PT as a benchmark, and, furthermore, we study GP-DC-PT to evaluate whether the two indicator sets can complement each other.

In Section V-A1, we initially compare GP-DC and GP-DC-PT with GP-PT across three key metrics: total return (TR), rate of return (RoR), and risk. This comparison spans 220 datasets, encompassing both 5 and 10-year periods. Notably, we avoid discussing the Sharpe ratio at this stage because including the same dataset twice (once for the 5-year period and once for the 10-year period) would render its interpretation inconclusive. We then proceed to Section V-A2, where we discuss each algorithm’s performance over distinct period lengths of 5 and 10 years. In this section, we provide comprehensive results and insights across all four metrics:

TABLE 5. Summary statistics of the GP-based algorithms across 220 datasets. We use boldface for the best values for each metric.

Measure	Algorithm		
	Total return		
	GP-DC	GP-PT	GP-DC-PT
Average	14.90%	12.20%	12.26%
Median	10.19%	6.08%	9.90%
Maximum	187.98%	571.21%	186.61%
Minimum	-149.58%	-57.50%	-229.56%
	Rate of return		
	GP-DC	GP-PT	GP-DC-PT
Average	1.43%	1.53%	1.02%
Median	1.08%	0.81%	1.02%
Maximum	27.00%	46.06%	7.74%
Minimum	-7.55%	-22.31%	-3.82%
	Risk		
	GP-DC	GP-PT	GP-DC-PT
Average	0.09	0.10	0.06
Median	0.07	0.07	0.05
Maximum	0.48	0.52	0.24
Minimum	0.01	0.01	0.00

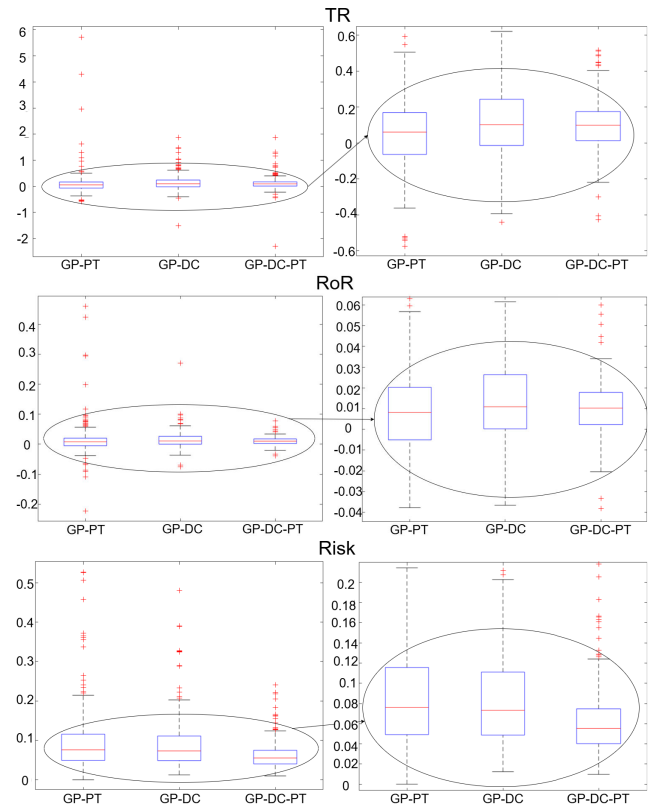
TR, RoR, risk, and the Sharpe ratio. Lastly, we compare the three GP algorithms in terms of their best models, in Section V-A3. Once the above comparisons are complete, we identify the best GP algorithm and compare it against the benchmarks of technical indicators (Section V-B) and buy-and-hold (Section V-C). Furthermore, we performed statistical analysis, including the non-parametric Kolmogorov-Smirnov (KS) statistical test, to assess the importance of our findings. As control algorithms, we used the algorithms that performed either best or worst on at least one metric. As an illustration, in Table 6, we used GP-PT as the control algorithm because it performed worst in terms of TR and risk, and in Table 9, we used GP-DC as the control algorithm because it has the best TR for the 5-year datasets.

A. COMPARING THE GP-BASED ALGORITHMS

1) OVERALL COMPARISON FOR THE THREE GP-BASED ALGORITHMS

We present, in Table 5, the summary statistics obtained from all 220 datasets for the metrics of TR, RoR, and risk. In terms of TR over the test set period, we have that both DC versions (GP-DC, GP-DC-PT) have higher average and median values than GP-PT. In addition, GP-DC has the highest average (14.90%) and median (10.19%) TR across the three algorithms. With regards to the RoR (i.e. average return per trade), we can observe that GP-PT has the highest value (1.53%), however, when it comes to median values both GP-DC and GP-DC-PT have higher values. This can be explained by the bigger range of results that are observed for GP-PT, which as a result, inflate its average value. Lastly, with regard to risk, both DC-based algorithms have lower values than GP-PT, with GP-DC-PT having the lowest values overall.

We can confirm the above results by studying also the distribution of results as presented in Figure 3, where the values of TR, RoR, and risk across 220 datasets for GP-DC, GP-PT, and GP-DC-PT are reported. We can observe that both GP-DC and GP-DC-PT algorithms outperform the GP-PT algorithm in the majority of cases. In contrast, the GP-PT algorithm has more outliers that correspond to massive profits or losses. This is consistent with the results on risk, where the GP-DC algorithm has a lower risk than the GP-PT algorithm in most cases. Moreover, GP-DC-PT further reduces risk by sacrificing TR and RoR, compared to both GP-DC and GP-PT algorithms.

**FIGURE 3.** Box plots for the three GP-based algorithms for TR, RoR, and risk.

We used the KS test to enrich the analysis of the three algorithms. Given that GP-PT is the benchmark, we compare it pairwise against GP-DC and against GP-DC-PT. The null hypothesis is that both distributions in each pair come from the same continuous distribution. To take the multiple comparisons (two in total) into account, we adjust the p -value by using the Bonferroni correction. Thus, at the 5% significance level, the null hypothesis is rejected for a p -value = $0.05/2 = 0.025$.

Statistical test results are shown in Table 6. In terms of TR, the p -value between GP-PT and GP-DC is 0.0125, thus below the 0.025 level, which suggests that the null hypothesis is rejected at the 5% significance level. There are similar results between GP-PT and GP-DC-PT for both TR and RoR.

This thus means that both GP-DC and GP-DC-PT are able to statistically outperform GP-PT in terms of TR; in addition, GP-DC-PT is also able to outperform GP-PT in terms of RoR. On the contrary, there are no statistically significant differences between GP-PT and GP-DC in terms of RoR, as well as in terms of risk for both pairwise comparisons. However, we note that GP-DC-PT's risk performance is only marginally not statistically significant at the 10% level (adjusted p -value threshold is 0.05, and the test's p -value was 0.0572).

TABLE 6. The p -values between GP-PT against each of GP-DC and GP-DC-PT algorithms; p -values below 0.025 appear in boldface to indicate statistical significance at the 5% level.

Algorithm	Total return	Rate of return	Risk
GP-DC	0.0125	0.1366	0.7512
GP-DC-PT	0.0066	0.0048	0.0572

The above results show that the proposed DC-based algorithms could be a complementary approach to the physical time method. GP-DC performs significantly better than GP-PT in terms of TR, while the risk is further reduced by combining both DC and technical indicators. Besides, both DC algorithms have less volatile results than the GP-PT algorithm. We will further compare their results from different angles in the following sections.

2) PERIODS

Recall that our datasets were created over two periods, i.e. for 5 and 10 years, respectively. In this section, we will consider the influence of the period on the performance of the three GP-based algorithms. In addition to TR, RoR, and risk, we now also place particular emphasis on the portfolio Sharpe ratio, which is now included alongside the above metrics, as key evaluation metrics for GP-DC, GP-PT, and GP-DC-PT.

As we can observe in Table 7, our proposed algorithms are doing very well in the 5-year period, particularly in terms of TR and RoR. GP-DC has the highest average (13.89%) and median (9.63%) values for both metrics, while GP-DC-PT comes second. With regards to risk, GP-DC-PT has again the lowest average (0.06) and median (0.05) values, while GP-PT comes second and GP-DC marginally third. When looking at the 10-year data, we find that GP-PT has the highest average TR (21.38%), which is probably affected by the outliers (e.g. maximum value 571.21%).¹ This can also be confirmed by looking at Figure 5, where we can clearly see a large number of big outliers for TR. On the other hand, GP-DC has the highest median TR (10.96%). In terms of RoR, GP-PT has the best average (2.47%) and median (1.00%)

¹Such outliers can be explained by a combination of a heavily bull market and a very passive trading strategy. For example, a particular GP model might recommend a buy action at the beginning of the 2-year test set period (for the 10-year dataset), and then sell towards the end of the test set period. Assuming that the respective dataset's prices were in a strong upwards trend, such an action can lead to a very profitable trade.

TABLE 7. Summary statistics of the GP-based algorithms on 5 and 10-year periods. We use boldface for the best values for each metric.

Period	Measure	Algorithm		
		Total return		
		GP-DC	GP-PT	GP-DC-PT
5 year	Average	13.89%	2.82%	10.23%
	Median	9.63%	3.83%	9.02%
	Maximum	187.98%	130.44%	130.98%
	Minimum	-33.08%	-57.50%	-40.45%
10 year	Average	15.91%	21.38%	14.29%
	Median	10.96%	10.13%	10.85%
	Maximum	147.52%	571.21%	186.61%
	Minimum	-149.59%	-53.59%	-229.56%
		Rate of return		
		GP-DC	GP-PT	GP-DC-PT
5 year	Average	1.58%	0.58%	1.13%
	Median	1.37%	0.61%	1.17%
	Maximum	10.07%	9.08%	7.74%
	Minimum	-7.55%	-10.93%	-3.82%
10 year	Average	1.28%	2.47%	0.91%
	Median	0.89%	1.00%	0.77%
	Maximum	27.00%	46.06%	6.00%
	Minimum	-2.83%	-22.31%	-3.33%
		Risk		
		GP-DC	GP-PT	GP-DC-PT
5 year	Average	0.10	0.09	0.06
	Median	0.08	0.07	0.05
	Maximum	0.48	0.45	0.22
	Minimum	0.01	0.00	0.01
10 year	Average	0.08	0.10	0.06
	Median	0.07	0.07	0.05
	Maximum	0.39	0.53	0.24
	Minimum	0.01	0.01	0.01

TABLE 8. Sharpe ratio of the GP-based algorithms on 5 and 10-year periods. We use boldface for the best values for each period.

Period	GP-DC	GP-PT	GP-DC-PT
5 year	0.5451	0.1867	0.7214
10 year	0.4266	0.3100	0.7081

values. Furthermore, in terms of risk, GP-DC-PT is again showing strong performance, as it has the lowest values, with GP-DC coming second.

Finally, for a comprehensive evaluation of each algorithm's risk-adjusted return performance, we establish portfolios for each GP variant (GP-DC, GP-PT, GP-DC-PT) within both the 5 and 10-year time frames. This results in the creation of six distinct portfolios, each comprising 110 datasets, with an equal allocation weight per dataset.² This methodology allows us to gain a holistic perspective on the performance of each algorithm across different time periods. Thus, in Table 8, we present the Sharpe ratio performance of each GP's portfolio. It is clear that the ranking remained consistent

²It's worth noting that this paper primarily centers on algorithmic trading, and the optimisation of portfolio weights falls beyond its current scope. Therefore, for simplicity, equal weights have been assigned. Future research will look into the optimisation of portfolio allocations.

for both the 5-year and 10-year periods, with GP-DC-PT achieving the top result, followed by GP-DC in second place, and GP-PT trailing behind in third. It reveals the consistent profit-generating capability of GP-DC-PT.

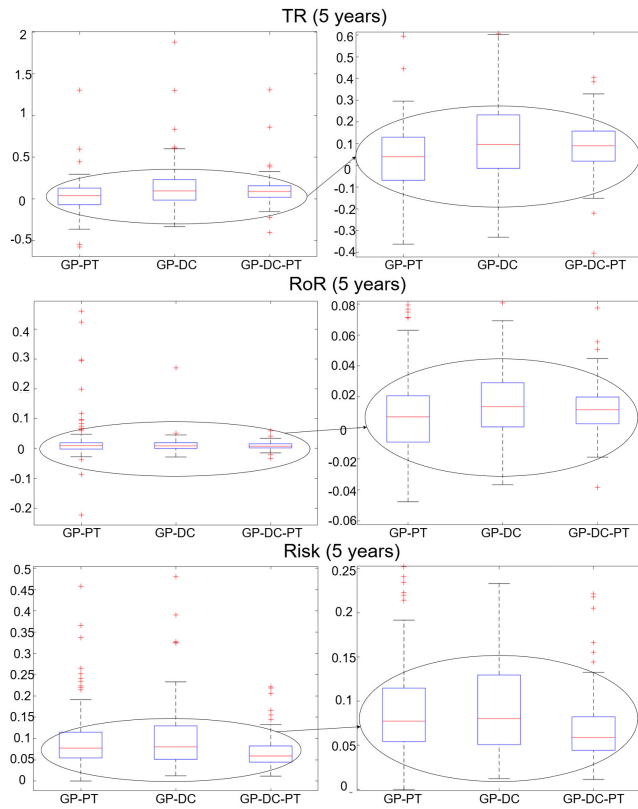


FIGURE 4. Box plot of summary statistics for GP-based algorithms under 5-year datasets for TR, RoR, and risk.

The above results are also confirmed by Figures 4 and 5, where we can observe that the GP-DC-PT algorithm outperforms both GP-DC and GP-PT concerning risk in both 5-year and 10-year datasets, while its TR and RoR are comparable to those of GP-DC. Elaborating on risk, we can see that, on the one hand, GP-DC-PT has a thinner box plot, indicating its lower risk in terms of 5-year datasets, and outperforms the GP-PT and GP-DC algorithms for this setting. On the other hand, by looking at Figure 5, we can observe that the GP-PT algorithm achieves several extreme values in several datasets, while the plots of GP-DC and GP-DC-PT algorithms are flat; the latter algorithms, hence, lead to lower risk.

Again, we use the non-parametric KS test to evaluate the above results. Table 9 presents the p -values of the KS test between the GP-DC algorithm (control algorithm, given that it had the best TR and RoR results) and each of the GP-PT and GP-DC-PT algorithms in terms of TR, RoR, and risk; p -values indicating statistical significance appear in boldface.

We remark that the distribution of GP-DC and GP-PT algorithms is statistically and significantly different in terms of TR and RoR in the 5-year datasets. Combined with

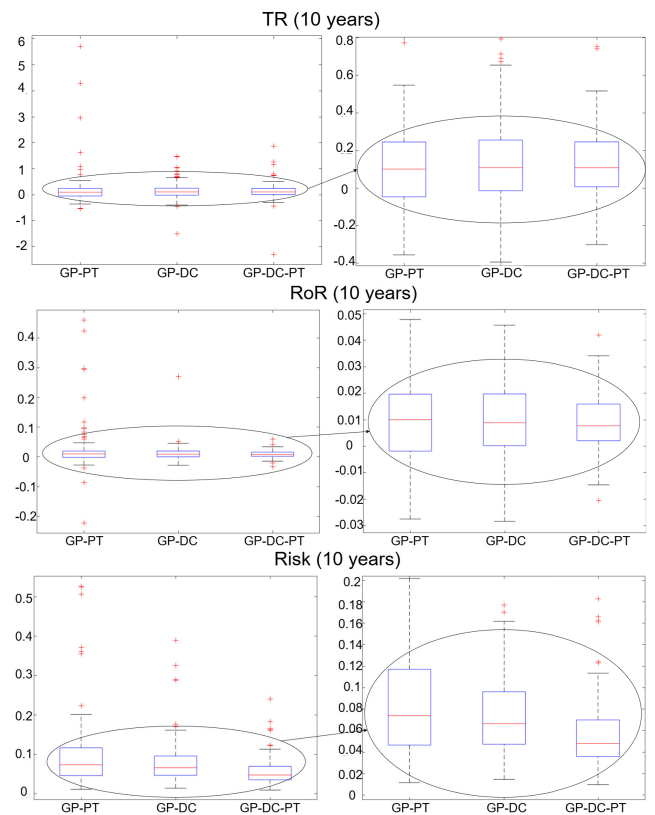


FIGURE 5. Box plot of summary statistics for GP-based algorithms under 10 years datasets for TR, RoR, and risk.

Table 7, we can conclude that GP-DC outperforms, in a statistically significant manner, the GP-PT algorithm in terms of TR and RoR for these datasets. Meanwhile, the difference between GP-DC and GP-DC-PT is also significant in terms of 5-year and 10-year risk. This further shows a considerable improvement in risk for the GP-DC-PT algorithm. For the rest of the comparisons, the differences in performance are not statistically significant.

TABLE 9. The p -values between GP-DC algorithm and each of the GP-PT and GP-DC-PT algorithms; p -values below 0.025 appear in boldface to indicate statistical significance at the 5% level.

	Total return	
	GP-PT	GP-DC-PT
5 year	0.0034	0.2395
10 year	0.5060	0.6186
	Rate of return	
	GP-PT	GP-DC-PT
5 year	0.0137	0.0055
10 year	0.5060	0.2395
	Risk	
	GP-PT	GP-DC-PT
5 year	0.5060	6.9784e-05
10 year	0.2395	7.2247e-04

In conclusion, the GP-DC algorithm performs better than the GP-PT algorithm in the short term (5 years). In contrast,

the GP-PT algorithm outperforms the GP-DC algorithm over the long term (10 years), whereas the GP-DC and GP-PT algorithms exhibit statistically significant differences in TR and RoR in 5 years (both TR and RoR are higher for GP-DC). Moreover, the GP-DC-PT algorithm significantly reduces the risk with a similar distribution of TR to the GP-DC algorithm in both 5- and 10-year datasets. In addition, although average values are worse than the other two algorithms, the box plot of the GP-DC-PT algorithm is rather flat and reflects stability; this is also indicated by the higher Sharpe ratio. Apart from providing further evidence that DC-based algorithms could complement physical time methods, this demonstrates the advantage of the GP-DC-PT algorithm over the GP-DC algorithm.

3) BEST MODEL

The previous sections provided average results, which are essential in the field of machine learning, giving us an understanding of how a particular algorithm is expected to perform. Here, we focus on presenting the top-performing results for each genetic programming (GP) algorithm. When we mention “best results”, we are referring to the *best tree* with respect to fitness, selected from 50 runs conducted on the training dataset. This selected tree is then applied to the test dataset, representing the best outcome from those 50 runs. This approach holds particular importance in the financial sector. For example, if an investor were to use a GP algorithm for stock market analysis, they would usually execute the algorithm multiple times and choose the best-performing tree to form their trading strategy. Hence, having an algorithm that excels in terms of the *best tree* is a crucial consideration in the financial sector. The specific results are presented in Table 10.

From Table 10, we can observe that GP-DC-PT has the best average and median values in both TR and RoR; note that the median TR of GP-DC-PT is three times higher than GP-DC's, which ranks second. We remark, however, that GP-DC-PT attains those results at the expense of slightly higher risk.

From Figure 6, the box plots show evidence of GP-DC-PT's better performance. In both box plots of TR and RoR, GP-DC-PT has a higher upper line and average line together with the same lower line. Regarding risk, the box plot shows that GP-DC-PT has a narrower range, but there is no evidence about which one is better.

To further analyse the statistical result, we again use the KS test. Since the best-performing algorithm is GP-DC-PT, we use it as the control algorithm. The results are shown in Table 11. Observe that the difference in all three metrics between GP-DC-PT and GP-PT is significant. At the same time, only the TR is significantly different between GP-DC-PT and GP-DC, as well as risk at the 10% level (adjusted p -value 0.05).

We can thus establish that GP-DC-PT outperforms GP-PT in TR and RoR. In contrast, GP-PT outperforms GP-DC-PT in risk. Besides, GP-DC-PT also outperforms GP-DC in terms of TR. For the rest of the comparisons, there are no statistically significant differences.

TABLE 10. Summary statistics of the best trees for GP-DC, GP-PT, and GP-DC-PT; best trees are those with the best fitness function value over 50 runs on the training dataset. We use boldface for the best values for each metric. Note that 0.00 risk indicates no trades happened.

Measurement	Algorithm		
	Total return		
	GP-DC	GP-PT	GP-DC-PT
Average	10.71%	7.43%	18.38%
Median	3.34%	1.41%	11.63%
Maximum	304.01%	234.70%	434.50%
Minimum	-123.65%	-68.78%	-465.70%
	Rate of return		
	GP-DC	GP-PT	GP-DC-PT
Average	0.88%	0.83%	1.62%
Median	0.75%	0.40%	1.27%
Maximum	11.73%	26.89%	19.65%
Minimum	-19.00%	-9.23%	-16.68%
	Risk		
	GP-DC	GP-PT	GP-DC-PT
Average	0.07	0.06	0.07
Median	0.05	0.04	0.05
Maximum	0.29	0.58	1.12
Minimum	0.00	0.00	0.00

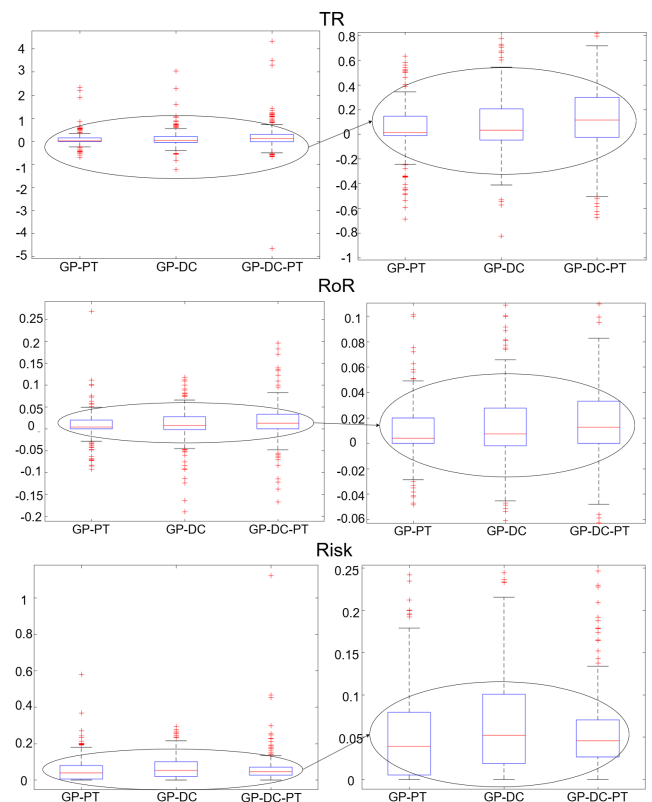


FIGURE 6. Box plot for the best trees of the GP-based algorithms.

In conclusion, in terms of best models, the GP-DC-PT algorithm outperforms the GP-DC and GP-PT algorithms for TR and RoR. However, GP-PT outperforms the GP-DC-PT algorithm in terms of risk. Nevertheless, when comparing

the significant improvements in terms of TR (from GP-PT's 1.41% to GP-DC-PT's 11.63%) to the very small increase in median risk (from GP-PT's 0.04 to GP-DC-PT's 0.05), we argue that GP-DC-PT is the best algorithm and should be selected for trading purposes.

TABLE 11. The p -values between GP-DC-PT algorithm and each of the GP-PT and GP-DC algorithms; p -values below 0.025 appear in boldface to indicate statistical significance at the 5% level.

Algorithm	Total return	Rate of return	Risk
GP-PT	1.0455e-04	0.0034	0.0024
GP-DC	0.0024	0.0857	0.0397

4) SUMMARY

By comparing the three GP-based algorithms' performance across all datasets, different periods, and best trees, we conclude that both DC-based algorithms outperformed GP-PT. This remark fulfils the objective of this paper, i.e. to demonstrate that DC-based trading strategies form a promising alternative to traditional physical time approaches. Furthermore, the GP-DC-PT algorithm has a significantly lower risk than the GP-DC algorithm. Meanwhile, although the GP-DC algorithm achieved better TR in many cases, the difference between GP-DC and GP-DC-PT algorithms is not statistically significant. Lastly, when analysing the Sharpe ratio results, it becomes evident that GP-DC-PT exhibits significantly higher values compared to GP-DC, with approximately a 33% improvement for the 5-year period and a remarkable 66% improvement for the 10-year period (refer to Table 8). The Sharpe ratio, being a comprehensive metric that incorporates both returns and risk, carries substantial significance, mirroring real-world behaviour by financial analysts and traders. Therefore, we emphasise the importance of placing greater weight on the insights derived from this metric. Consequently, we assert that, when it comes to striking a balance between return and risk, GP-DC-PT emerges as the preferred choice among the three GP-based algorithms. As a result, the subsequent sections will involve benchmarking the GP-DC-PT algorithm against technical indicators and a traditional buy-and-hold trading strategy.

B. COMPARISON OF GP-DC-PT ALGORITHM TO TECHNICAL INDICATORS

To compare the effectiveness of our approach to traditional technical analysis, we benchmark the GP-DC-PT algorithm against three commonly used indicators: Moving Average Convergence Divergence (MACD), On Balance Volume (OBV), and Momentum (MTM), also used in [36]. The outcome of this comparison can be seen in Table 12, where a value of 0 means that the respective algorithm did not perform any trades. Note that GP-DC-PT exhibits a very high median TR, nearly three times the second-best value, together with a second-best average TR. In terms of RoR, GP-DC-PT has the second-best average value and the best median. With respect to risk, GP-DC-PT has the lowest average and median risk.

In addition, we calculate the Sharpe ratio of the portfolio, where each of the 110 datasets is given equal weight, for 5- and 10-year datasets separately in Table 13. The previous findings are fully reflected in the Sharpe ratio as well, where GP-DC-PT demonstrated a clear advantage, both in the 5-year and 10-year datasets, in comparison to the other three indicators.

TABLE 12. Summary statistics of the GP-DC-PT algorithm and technical analysis across 220 datasets. We use boldface for the best values for each measure.

Measure	Algorithm			
	Total return			
	GP-DC-PT	MACD	OBV	MTM
Average	12.26%	10.90%	2.78%	12.58%
Median	9.90%	1.15%	0.51%	3.92%
Maximum	186.61%	360.11%	152.31%	514.56%
Minimum	-229.56%	-97.24%	-177.72%	-112.67%
	Rate of return			
	GP-DC-PT	MACD	OBV	MTM
Average	1.02%	1.51%	0.37%	0.32%
Median	1.02%	0	0	0.10%
Maximum	7.74%	36.01%	35.65%	10.98%
Minimum	-3.82%	-24.31%	-24.60%	-2.59%
	Risk			
	GP-DC-PT	MACD	OBV	MTM
Average	0.06	0.11	0.08	0.14
Median	0.05	0.06	0.06	0.09
Maximum	0.24	1.58	0.76	1.82
Minimum	0	0	0	0.02

TABLE 13. Sharpe ratio of the GP-DC-PT algorithm and technical analysis. We use boldface for the best values for each period.

Period	GP-DC-PT	MACD	OBV	MTM
5 year	0.7214	0.1932	-0.0110	0.1711
10 year	0.7081	0.2620	0.1453	0.2910

We draw the box plot for the GP-DC-PT algorithm and each technical indicator to make a visual comparison. As shown in Figure 7, the upper line of the TR for each algorithm is similar. However, the GP-DC-PT algorithm displays a higher lower line closer to zero. It indicates that almost 75% datasets of the GP-DC-PT algorithm have positive TR. On the contrary, around 50% datasets of the three technical indicators lead to a negative TR. This is also the case for the RoR; both the average value and the majority cases of the GP-DC-PT algorithm's RoR are higher than those of the three technical indicators. Furthermore, the upper line of the risk for the GP-DC-PT algorithm is much lower than that of the technical indicators, indicating a distribution of less variability for GP-DC-PT. We argue, therefore, that the high average TR and RoR values observed in MACD and MTM, respectively, may be influenced by a few extreme data points.

Table 14 presents the KS test that confirms the above results. As we can observe, by considering also the

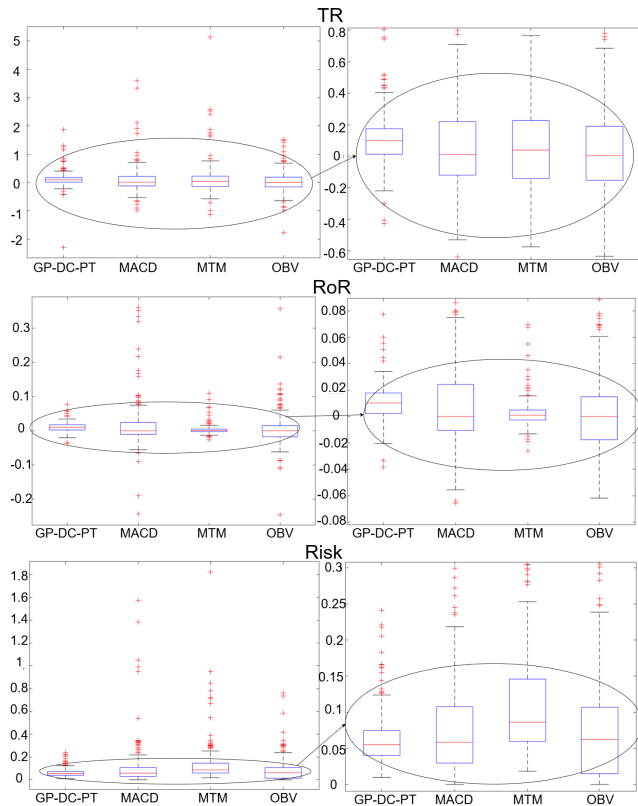


FIGURE 7. Box plot of TR, RoR, and risk for GP-DC-PT and technical indicators.

Bonferroni correction for the three comparisons, all p values below 0.0167 (i.e., $0.05/3$) reject the null hypothesis at the 5% significance level and indicate that the differences are statistically significant in terms of TR, RoR, and risk.

TABLE 14. The p -values between GP-DC-PT algorithm and technical analysis; p -values below $0.05/3 \approx 0.0167$ appear in boldface to indicate statistical significance at the 5% level.

Algorithm	Total return	Rate of return	Risk
MACD	4.35E-05	5.91E-08	6.78E-05
MTM	6.78E-05	1.49E-17	0.0092
OBV	5.91E-08	4.11E-12	1.04E-07

By the discussion above, we can conclude that our proposed algorithm, GP-DC-PT, has a significant advantage compared to the popular technical indicators. For some specific stocks, technical indicators perform well, with even better TR than the GP-DC-PT algorithm. Across the entire 220 datasets, however, the GP-DC-PT algorithm significantly outperforms the other three technical indicators at the 5% significance level.

C. COMPARISON TO THE BUY-AND-HOLD STRATEGY

We now compare GP-DC-PT with the state-of-the-art baseline approach, that is, the buy-and-hold strategy. It is worth noting that the buy-and-hold strategy only has one trade

during the whole period, where we buy one unit of stock on the first day of trading and we sell it on the last day; hence, the risk cannot be calculated. Furthermore, due to buy-and-hold making only a single trade (while GP-DC-PT makes several), it is fairer to compare them across the TR over the test set period rather than also using the RoR.

As we can observe from Table 15, the GP-DC-PT algorithm has the highest median TR. In contrast, the buy-and-hold strategy performs better on the rest of the metrics. It is worth noting that the buy-and-hold strategy achieves the highest maximum result (1753.05%), indicating that extreme values cause its high average effect. This is because some of the datasets come from a predominantly bull market (i.e., there is a strong upwards trend in the price series). This thus means that the last price can differ significantly from the first price (especially for the 10-year datasets, where the test set is 2 years long). For example, the above return of 1753.05% comes from NASDAQ's Plug Power Inc., which for the period of 2018-2020 saw its daily price skyrocket due to a combination of factors, such as the increasing interest in clean energy, significant partnerships and deals, and strong financial performance. As a result, such a high TR result can significantly affect the mean values, which are sensitive to outliers. To deal with this issue, we also analysed the results by only using all results within two standard deviations of the mean. Therefore, the average (standard deviation) TR of the buy-and-hold strategy, and GP-DC-PT algorithm becomes 12.13% (0.46) and 16.97% (0.33), respectively. Last but not least, as indicated in Table 16, GP-DC-PT outperforms the buy-and-hold strategy in terms of the Sharpe ratio when considering both 5-year and 10-year datasets. In this case, the GP-DC-PT algorithm performs better compared to the buy-and-hold strategy.

Furthermore, by looking at the box plot shown in Figure 8, we can observe that the lower line of the GP-DC-PT algorithm is above 0, which means the GP-DC-PT algorithm has a positive TR at 75% stocks. Besides, although the upper line of the buy-and-hold strategy is around 0.3, better than the GP-DC-PT algorithm, its lower line is around -0.1 . Furthermore, the several outliers of the buy-and-hold strategy make the previous high average TR less persuasive.

Even so, we need to look at the statistical result; as before, the KS test is used. The p -value between the GP-DC-PT algorithm and buy-and-hold strategy is 0.0169, below 0.05 (the 5% significance level), indicating the difference is statistically significant. Combined with the discussion above, we argue that the GP-DC-PT algorithm significantly outperforms the buy-and-hold strategy.

Lastly, it is worth noting that the GP-DC-PT algorithm could use the TR as the fitness function instead of the Sharpe ratio that we currently use. In this case, its average TR jumps to over 30%, which is higher than buy-and-hold's 27.21%.³

³Of course, since risk would no longer be part of the fitness function, the risk of GP-DC-PT would increase from 0.06 to 0.10. Nevertheless, this shows the flexibility of GP algorithms, which can easily adapt their fitness function to the different needs of traders.

TABLE 15. Summary statistics between GP-DC-PT algorithm with buy-and-hold across 220 datasets. We use boldface for the best values for each measure.

Measurement	Total return	
	GP-DC-PT	buy-and-hold
Average	12.27%	27.21%
Median	9.90%	6.25%
Maximum	186.62%	1753.05%
Minimum	-229.56%	-89.62%

TABLE 16. Sharpe ratio between GP-DC-PT algorithm with buy-and-hold. We use boldface for the best values for each period.

Period	GP-DC-PT	buy-and-hold
5 year	0.7214	0.1576
10 year	0.7081	0.2276

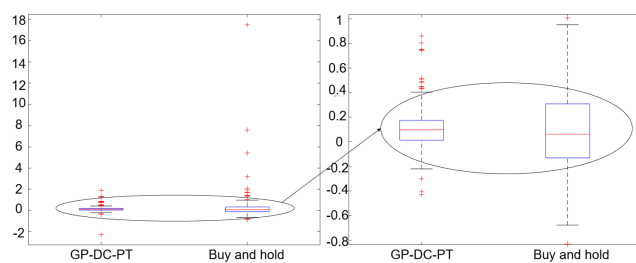


FIGURE 8. Box plot of TR between GP-DC-PT algorithm and buy-and-hold strategy.

VI. CONCLUDING REMARKS

Our study explored the advantages of using genetic programming together with indicators based on the framework of Directional Changes to create novel trading strategies. Our main achievement was conducting a thorough investigation of the performance of these trading strategies, carrying out experiments on 220 datasets from 10 different markets, covering time periods of 5 and 10 years.

Our investigation showed that both our proposed GP-DC and GP-DC-PT algorithms outperformed the baseline GP-PT in terms of total return (TR) and risk, with GP-DC-PT demonstrating the lowest overall risk across all tested datasets. When considering performance by timeframe, the 5-year period saw GP-DC leading in TR and Rate of Return (RoR), while maintaining lower risk compared to GP-DC-PT. However, GP-DC-PT proved its consistency by achieving the highest Sharpe ratio across both 5 and 10-year periods, even when GP-PT yielded the highest average TR in the 10-year analysis. Notably, even the best-performing models showed that GP-DC-PT could surpass both GP-DC and GP-PT in TR and RoR, albeit with slightly higher risk compared to GP-DC for these top models.

When GP-DC-PT was compared to technical analysis trading strategies, it significantly outperformed three commonly used technical indicators (MACD, OBV, MTM) in terms of return, risk, and the Sharpe ratio. GP-DC-PT also showed a better Sharpe ratio across both 5 and 10 year periods when

compared to buy-and-hold. The improvement in Sharpe ratio implies that GP-DC-PT is particularly suitable for portfolios where risk-adjusted performance is critical, such as in hedge funds or algorithmic trading desks that prioritise robust risk management.

These results highlight the practical and theoretical strength of combining GP with DC-based indicators. Our findings show that DC indicators, particularly when used alongside physical-time indicators, offer a more robust alternative to traditional time-based trading features. This supports further theoretical development in event-driven modelling and offers a new perspective for feature engineering in financial machine learning. The hybrid GP framework introduced in this study also contributes to a deeper understanding of modelling non-stationary financial time series using both event-based and time-based signals.

Despite the promising results, several limitations must be acknowledged. The backtesting results do not guarantee performance in live trading environments, where latency, slippage, and market impact may affect profitability. Moreover, our portfolio evaluation employed equal weighting, whereas real-world deployment would require asset allocation based on investor-specific constraints and objectives.

Future research will continue to explore the creation of novel trading strategies mixing technical analysis and DC indicators in different ways. In our current study, we directly used these indicators to train the GP algorithm. An interesting research direction is to employ a more specialized GP approach that separates the search for DC and physical time indicators. This could potentially lead to the development of more effective trading strategies. Additionally, alternative approaches, like the walk-forward method, for parameter tuning instead of dividing datasets into training, validation, and test sets are also worth exploring.

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